QUANT

Mensuration & Geometry

(Circles, Quadrilaterals & Trigonometry)

Assignment 04

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Geometry Online Assignment Cluster II -04

1. The angles of depression of the top and the bottom of a coconut tree as seen from the top of a tower 50 metres high, are 30° and 60° respectively. Find the height of the coconut tree.

1. 33.33 2. 66.66 3. 47.21 4. 98.41

2. The angle of depression of a point P on the bank of a river when seen from the roof of a building 17 m high is 60°. The angle of depression of point Q, on the opposite bank of the river when seen from the same roof is 30°. Find the width of the river.

1. 11.11 2. 22.22 3. 19.63 4. 34.5

3. Two aeroplanes, flying in opposite directions, are at the points A and B, 300 metres high from the ground. When the pilots look at a point P on the ground the angles of depression made by them are 30°and 60° respectively. Find the distance between the two aeroplanes.

1. 182.4 2. 178.6 3. 692.8 4. 793.4

4. A church and a temple are on either side of a road. When one sees from the roof of the church towards the top of the temple the angle of elevation is 30° and when one sees towards the foot of the temple, the angle of depression is 60°. If the height of the church is 8 metres, then find the height of the temple.

1. 10.66 2. 14.22 3. 15.33 4. 16.67

5. When one looks from the foot and the top of a tower at the roof of a building the angles of elevation and depression are of 27° and 63° respectively. If the height of the building is 20 metres, find the height of the tower. (tan 63° = 2)

1. 100m 2. 200m 3. 300m 4. 150m

6. There is a cliff 100 metres high on the bank of a lake. A hunter is sitting on the cliff. He looks at a cloud in the sky, and the angle of elevation is found to be of 30°. When he looks at the image of the cloud in the lake, the angle of depression is 60°. Find the height of the cloud from the lake surface.

1. 133.3 2. 266.4 3. 197.2 4. 412.8

7. An antenna is fitted on the top of a building, which is 12 metres tall. A boy is standing at some distance from the building. First he looks at the bird on the antenna and then at the top of the building. He finds that the angle of elevation of the bird is 36^0, whereas the angle of elevation of the top of the building is 32°. Find the distance of the boy from the building and the height of the antenna (tan 32° = 0.624, tan 36° = 0.726)

1. 2.12, 21.12 2. 9.1, 19.1 3. 2.3, 23.1 4. 1.96, 19.23

8. The walls of the building on both sides of a street are parallel to each other. A ladder 15 m long is placed so that its top just reaches the window of one building at the height 12 m. On turning the ladder over to the other side of the street, its top touches the other building at the height of 9 m. Find the breadth of the street.

1.16m 2. 21m 3. 23m 4. 29m

9. A point 'P' whose co-ordinate is $(-3,6)$ is in $X-Y$ plane. If the origin is shifted to the point $(5,8)$, then what will be the co-ordinate of point 'P' with respect to the new origin?

1. $(-7,-2)$ 2. $(-8,-2)$ 3. $(-2,5)$ 4. $(-9,-3)$

10. Find the equation of the straight line that passes through the points $(1,2)$ and $(7,8)$.

1. $x+y=1$ 2. $x-y=1$ 3. $x+y=0$ 4. $x-y+1=0$

Answer key & Explanations

Q. No.	Ans.	Exp.
1.	1	Let the height of the coconut tree = h and distance between coconut tree and tower = x. $\therefore$ after making the diagram it can be seen that also $\frac{50-h}{x} = \tan 30^0 \Rightarrow \frac{(50-h)\sqrt{3}}{50} = \frac{1}{\sqrt{3}}$. Solving this, we get h = 33.33 m.
2.	3	PQ will be the width of the river. Let the distance between point P and foot of the building be y. $\therefore$ after making the diagram, it can be seen that $\frac{17}{y} = \tan 60° \Rightarrow \frac{17}{y} = \sqrt{3} \Rightarrow y = 17/\sqrt{3}$. Also $\frac{17}{x+y} = \tan 30°$. $\therefore \frac{17}{x+\frac{17}{\sqrt{3}}} = \frac{1}{\sqrt{3}}$ $\Rightarrow 17\sqrt{3} = x + \frac{17}{\sqrt{3}}$. $\therefore x = 17\sqrt{3} - \frac{17}{\sqrt{3}}$ $\Rightarrow x = \frac{51-17}{\sqrt{3}} = \frac{34}{\sqrt{3}} = 19.63m$.
3.	3	Let the total distance between 2 aero planes be $x + y$, where x is distance between point P and point below A on the ground and y is the distance between point P and point below B on the ground. $\therefore \frac{300}{x} = \tan 30^\theta \Rightarrow \frac{300}{x} = \frac{1}{\sqrt{3}}$ $\Rightarrow x = 300\sqrt{3}$ and $\frac{300}{y} = \tan 60^\theta \Rightarrow \frac{300}{y} = \sqrt{3} \Rightarrow y = 300/\sqrt{3}$. $\therefore x + y$ = 519.6 + 173.2 = 692.8m.
4.	1	Let the height of the temple = H. H = 8 + h and distance between church and temple = x. Height of the church = 8 m. So from the diagram, it can be seen that $\frac{8}{x} = \tan 60°$ $\Rightarrow \frac{8}{x} = \sqrt{3}$ $\Rightarrow x = \frac{8}{\sqrt{3}}$. Also $\frac{h}{x} = \tan 30^\theta$ $\Rightarrow \frac{h}{x} = \frac{1}{\sqrt{3}} \Rightarrow h = \frac{x}{\sqrt{3}}$. $\therefore h = 8/3$. Hence total height of temple = 8 + 8/3 = 10.66 m
5.	1	Let total height of tower = H. $\therefore H = h + 20$ where h is height of the tower above the building. Let x be the horizontal distance between the tower and the building. $\therefore$ as per diagram, $\frac{x}{20} = Cot\, 27 \Rightarrow \frac{x}{20} = \tan 63$. $\therefore x / 20 = 2 \Rightarrow x = 40$. Also $h/x = \tan 63^0$ $\Rightarrow h/40 = 2$ $\Rightarrow h$ = 80m. $\therefore$ height of the tower = 80 + 20 = 100 m.
6.	1	Let the lake surface be at a horizontal distance = x. Let the height of the cloud = H. Also $H = h + 100$ where h is the height of cloud above the cliff.

		$\frac{100}{x} = \tan 60^0 \Rightarrow \frac{100}{x} = \sqrt{3}$. $\therefore x = \frac{100}{\sqrt{3}}$. Also $h/x = \tan 30^0$ $\Rightarrow \frac{h\sqrt{3}}{100} = \frac{1}{\sqrt{3}} \Rightarrow h = \frac{100}{3}$. $\therefore$ Total height of the cloud $= 100 + \frac{100}{3} = \frac{400}{3} = 133.33$ m.
7.	4	Let distance between the boy and the building = x and h is the height of the antenna. $\therefore \frac{12}{x} = \tan 32^0 \Rightarrow \frac{12}{x} = 0.624 \Rightarrow x = 19.23m.$ Also $\frac{h+12}{x} = \tan 36 \Rightarrow \frac{h+12}{19.23} = 0.726$. Solving this we get h = 1.96 m.
8.	2	Let P be the point on the ground where the 2 ladders meet. Let A and B be the feet of the 2 buildings of 9m and 12m height respectively. Required distance = PA + PB. $\therefore PA^2 = 15^2 - 9^2 = 225 - 81$ $= 144.$ $\therefore PA = 12$ m and $PB^2 = 15^2 - 12^2 = 225 - 144 = 81.$ $\therefore PB = 9$ m. Hence breadth of the street i.e. PA + PB = 12 + 9 = 21 m.
9.	2	Co-ordinate of 'P' with respect to new origin is $x = x - x'$ and $y = y - y'$ $i.e. x = -3 - 5$ and $y = 6 - 8$ Or $x = -8$ and $y = -2$ Hence, the co-ordinate of 'P w.r.t new origin is $(-8, -2)$
10.	4	We get $y - 8 = \frac{8-2}{7-1}(x-7)$ or $x - y + 1 = 0$